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(54) **TOOL TRACKING DEVICE WITH
MULTIPLE AND APPLICATION SETTABLE
BEACON TRANSMISSION RATES**

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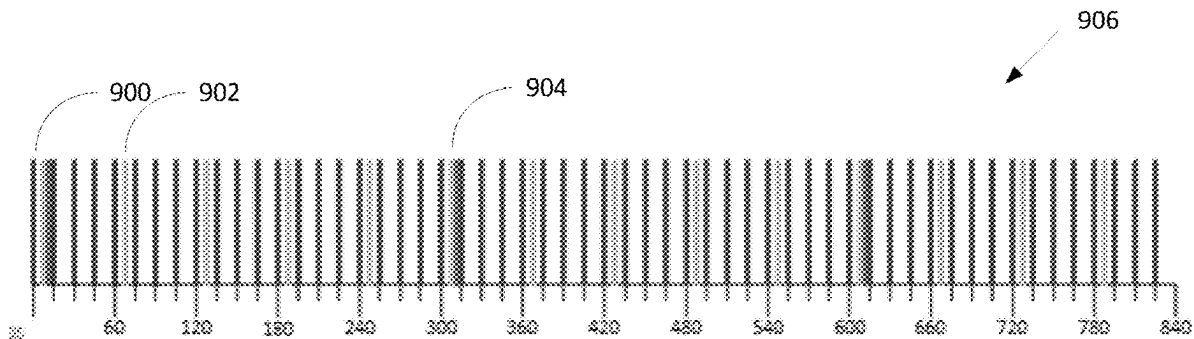
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(57) **ABSTRACT**

A method for modifying a repeat interval of a locator beacon is described. The method includes transmitting repeatedly, by a beacon transmitter, a first advertising beacon signal through a first number of transmission repetitions spaced at a first repeat interval. The method further includes transmitting repeatedly, by the beacon transmitter, a second advertising beacon signal through a second number of transmission repetitions at a second repeat interval, wherein the second advertising beacon signal includes an identifier that provides identification of the beacon transmitter and user information. The method further includes receiving a modification instruction from a user device. The modification instruction is configured to modify the first repeat interval into a third repeat interval. The method also includes transmitting repeatedly, by a beacon transmitter, the first beacon signal through the first number of transmission repetitions spaced at the third repeat interval.



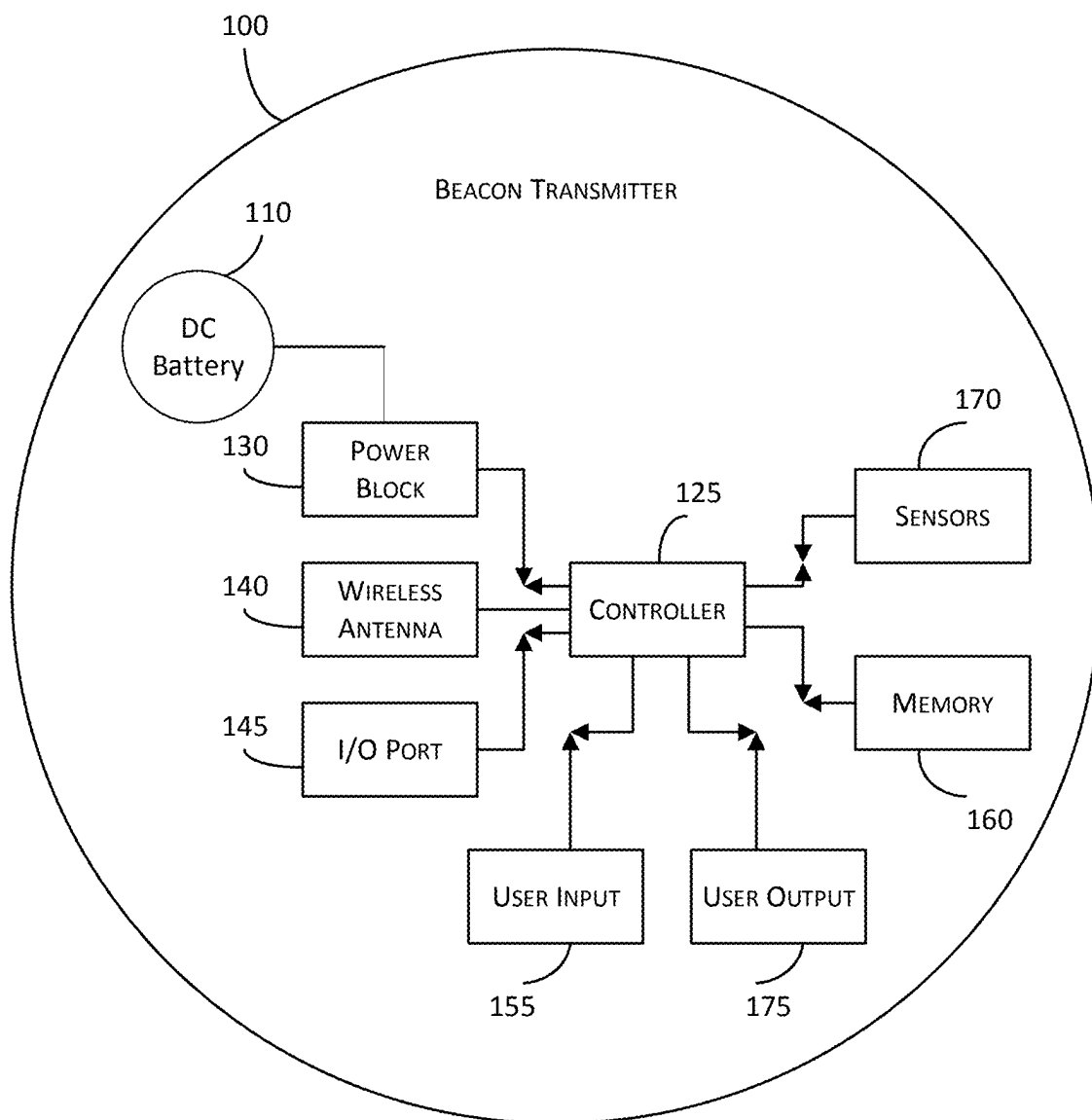


FIG. 1A

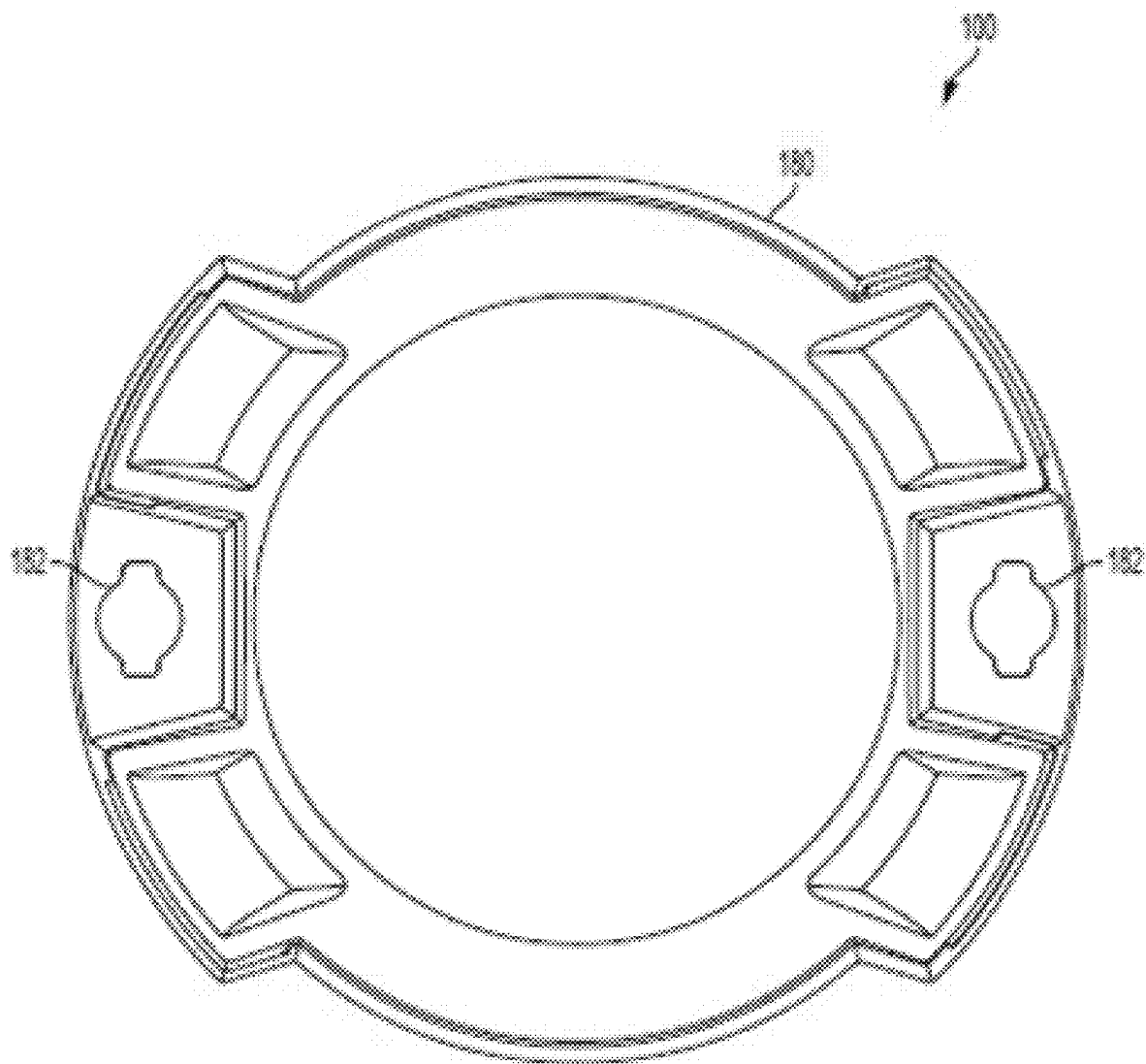


FIG. 1B

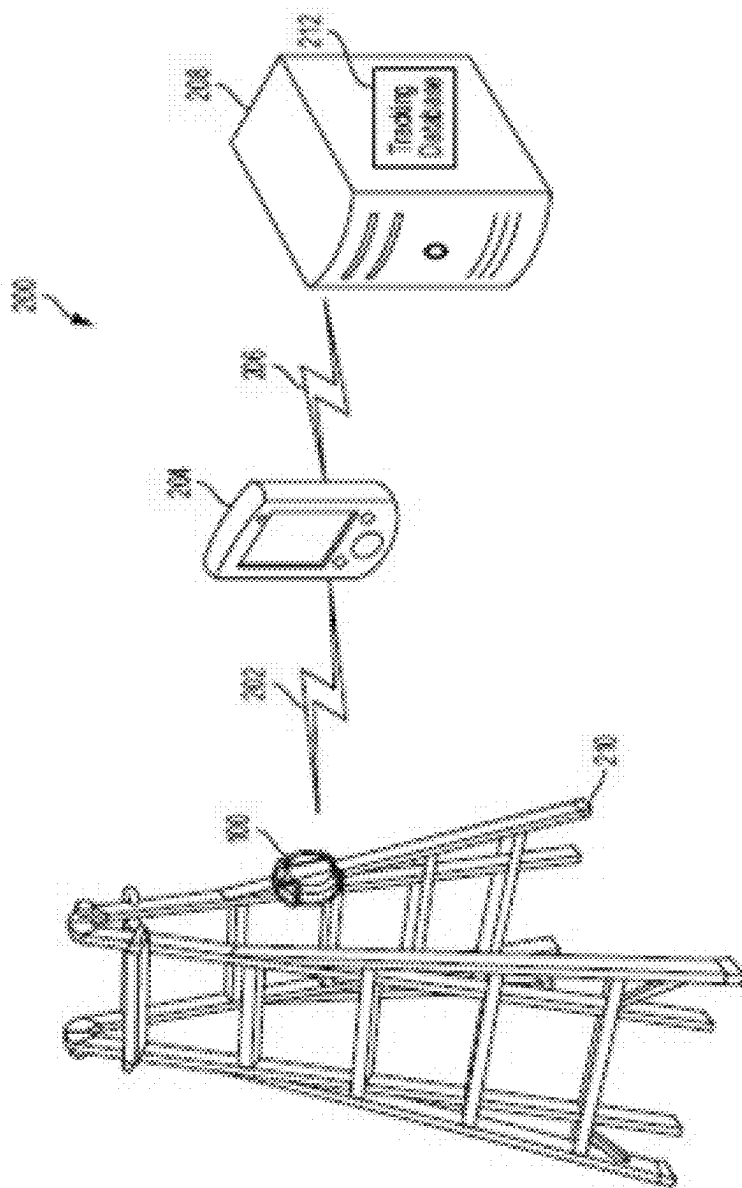


FIG. 2

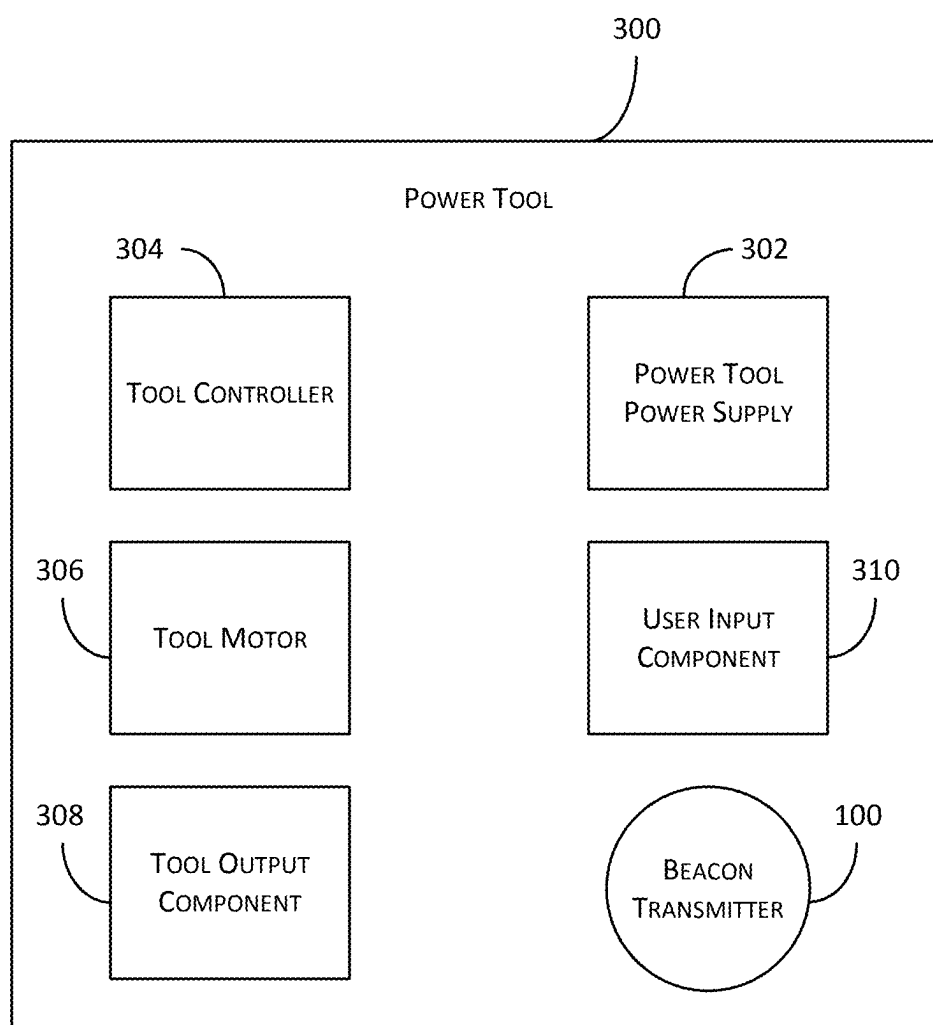


FIG. 3A

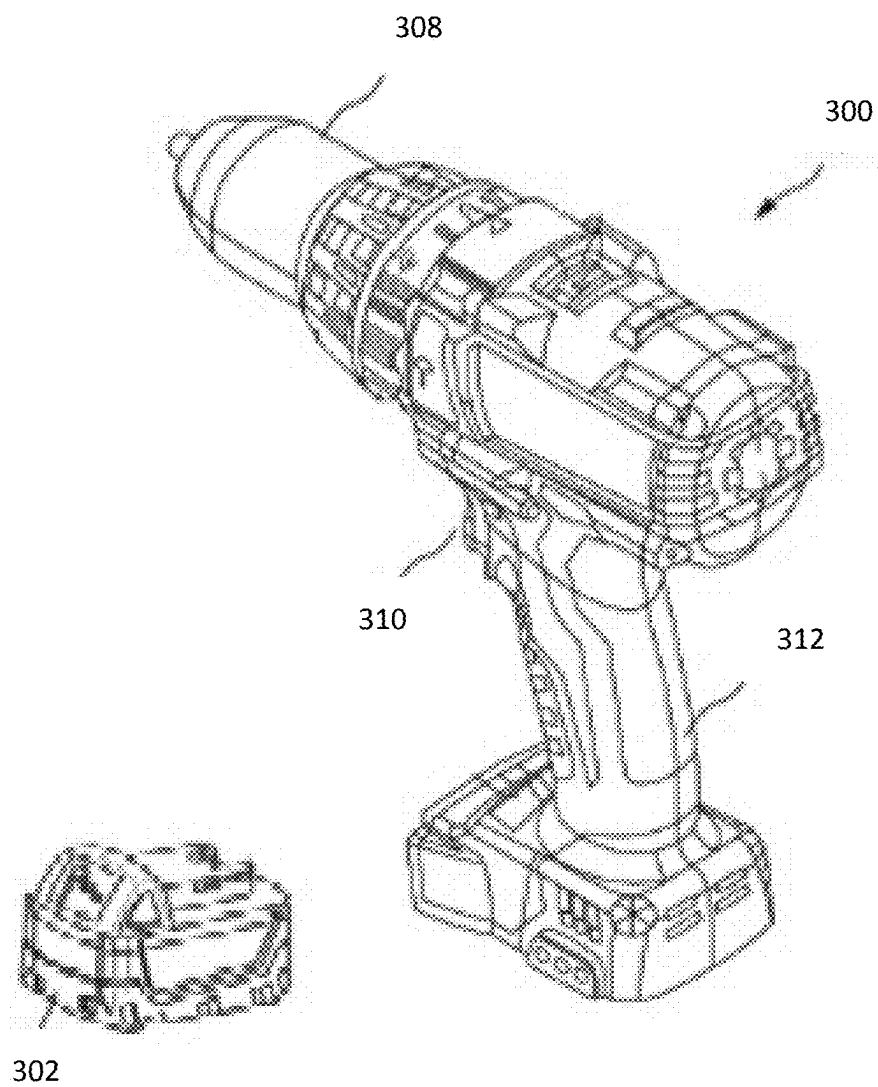


FIG. 3B

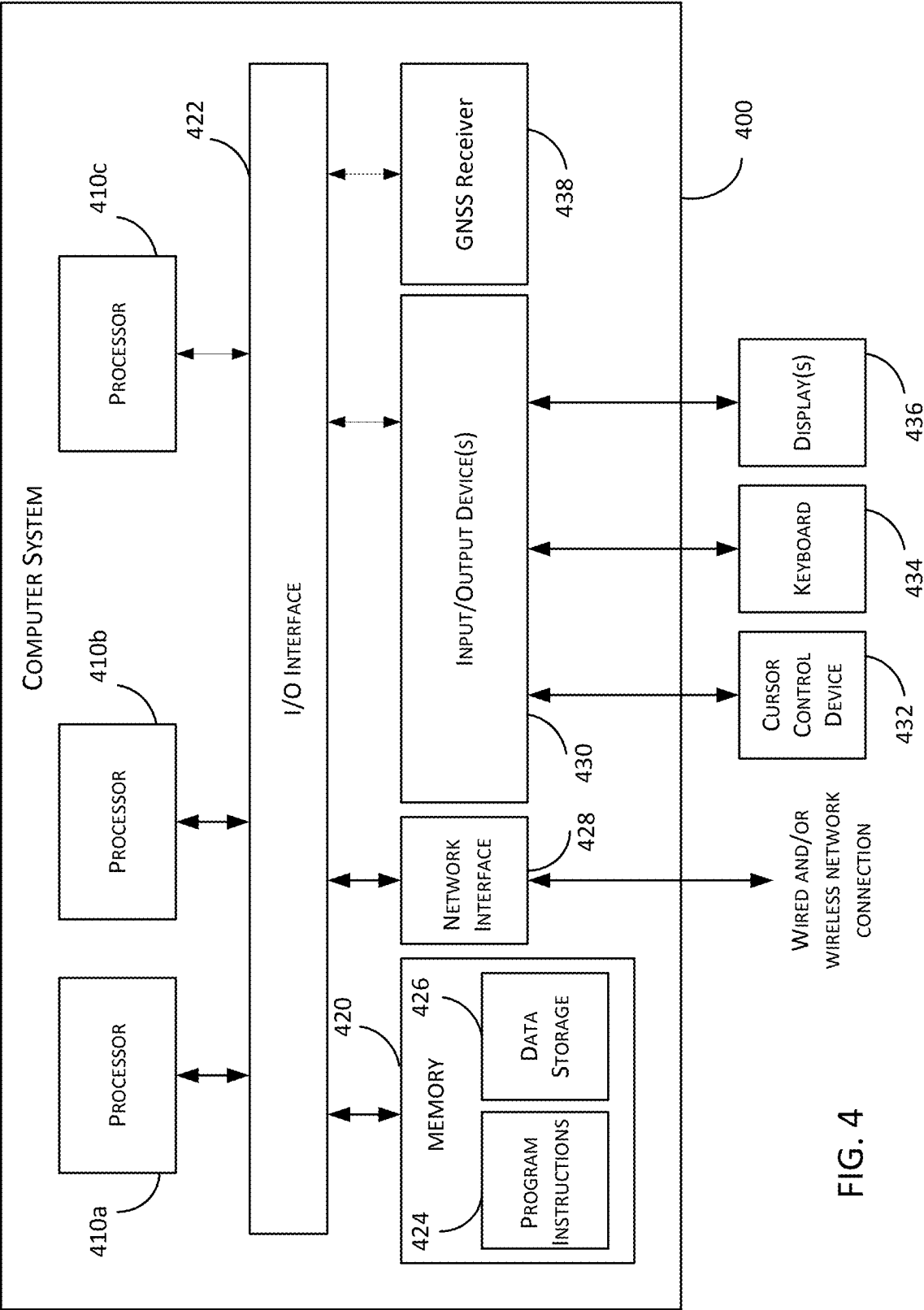


FIG. 4

500

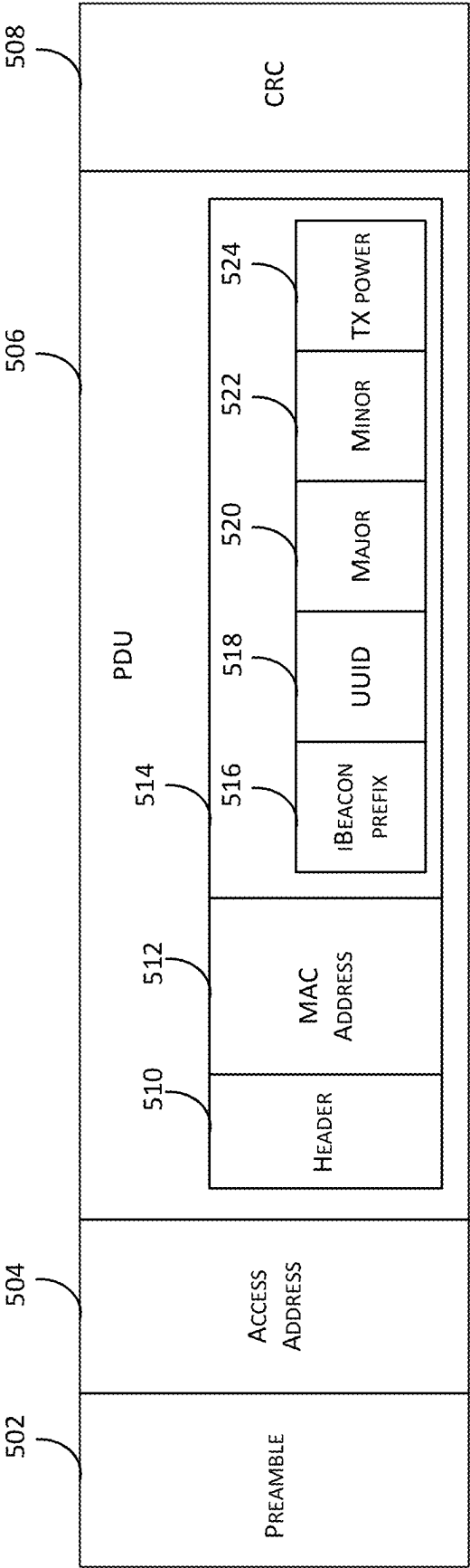


FIG. 5

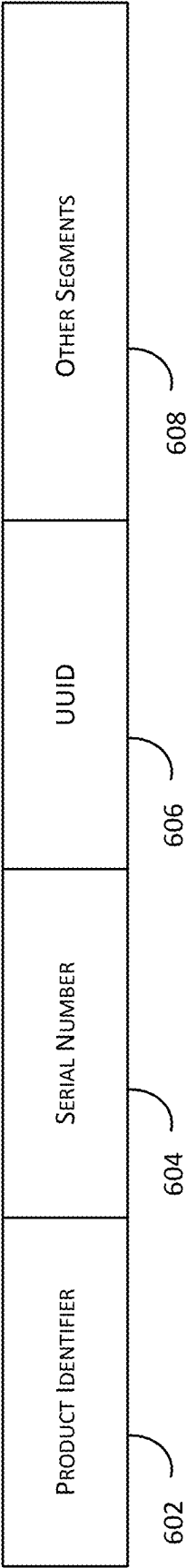


FIG. 6

FIG. 7

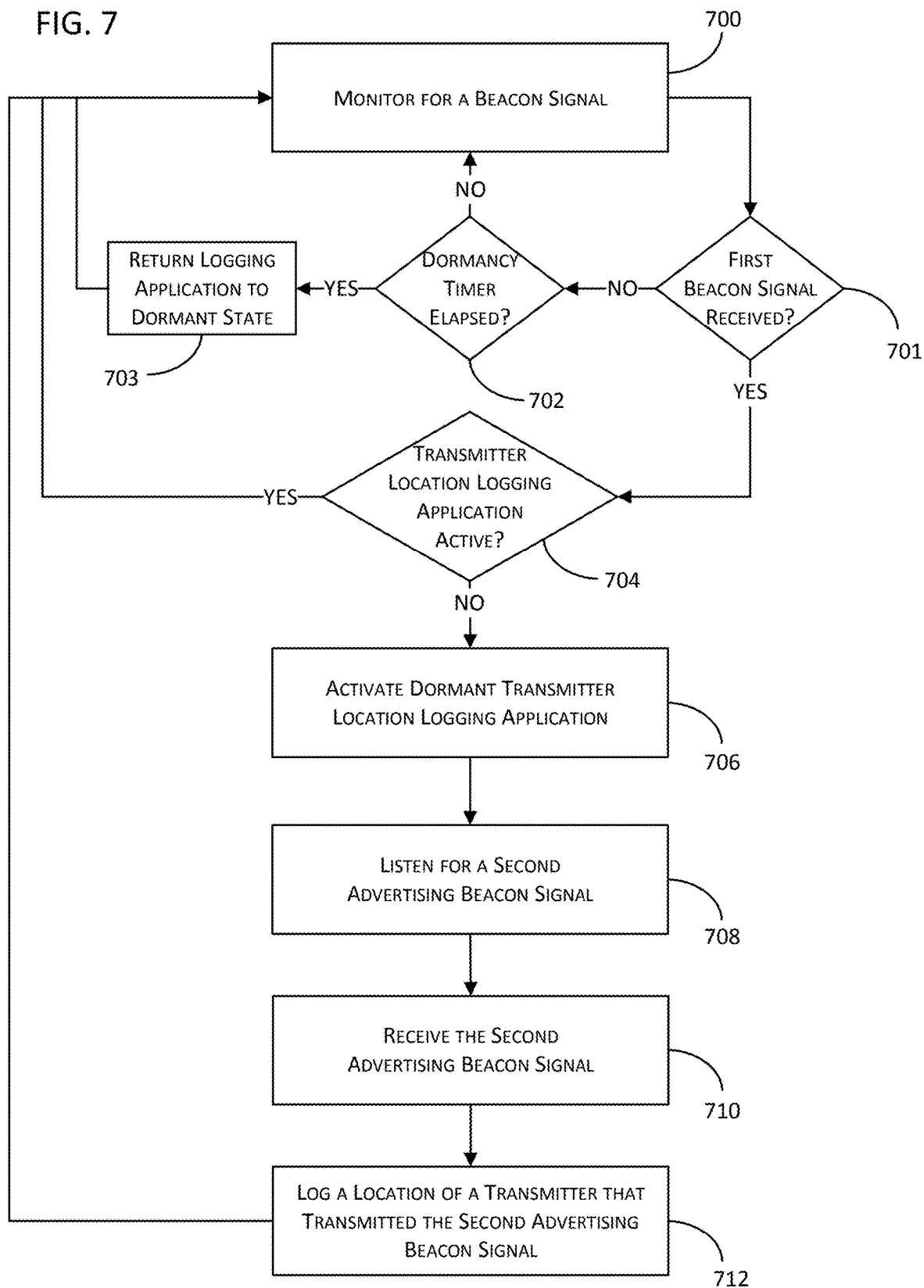
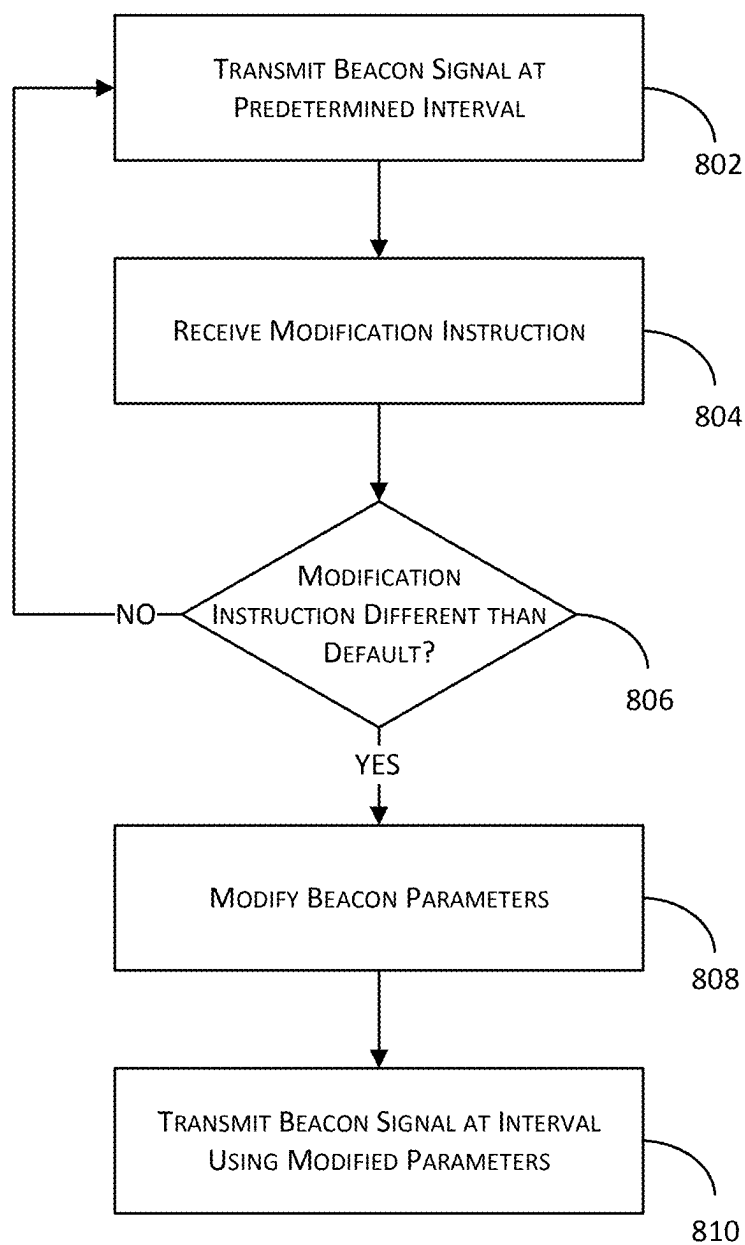


FIG. 8

800



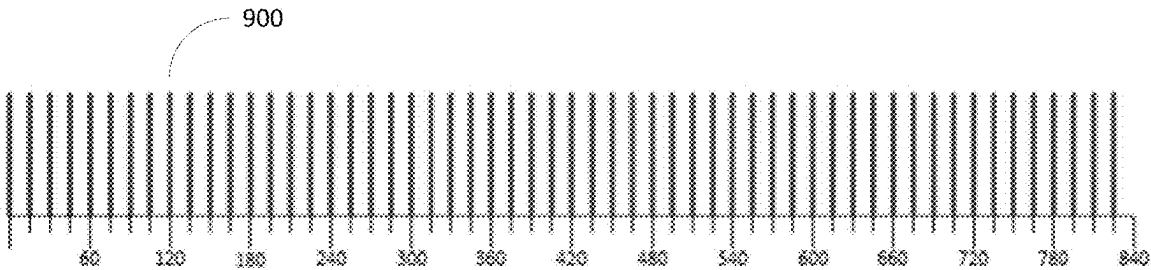


FIG. 9A

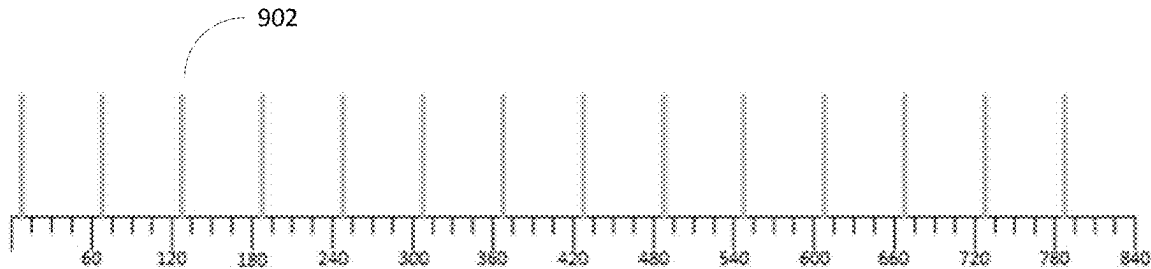


FIG. 9B

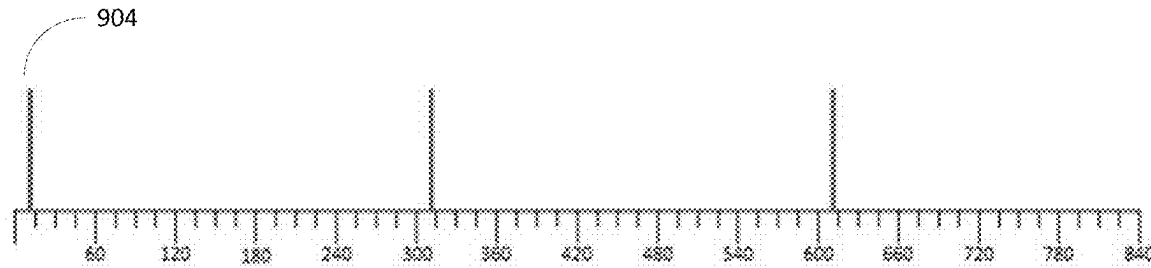


FIG. 9C

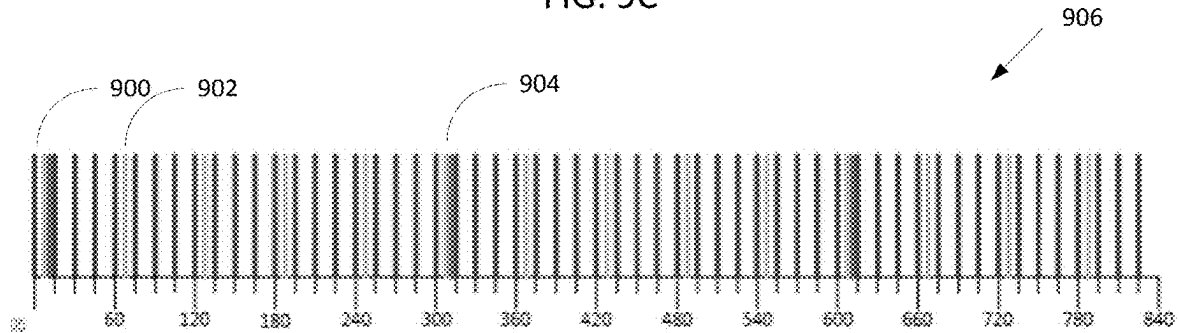


FIG. 9D

**TOOL TRACKING DEVICE WITH
MULTIPLE AND APPLICATION SETTABLE
BEACON TRANSMISSION RATES**

RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/416,401, filed Jan. 18, 2024, which is a continuation of U.S. patent application Ser. No. 17/051,502, filed Oct. 29, 2020, which is a national phase filing under 35 U.S.C. § 371 of International Application No. PCT/US2020/046105, filed Aug. 13, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/886,124, filed Aug. 13, 2019, the entire content of each of which is hereby incorporated by reference.

FIELD

[0002] The present technology relates to communication and logging of location for tools and other devices.

SUMMARY

[0003] In some embodiments a method for modifying a repeat interval of a locator beacon is described. The method includes transmitting repeatedly, by a beacon transmitter, a first advertising beacon signal through a first number of transmission repetitions spaced at a first repeat interval. The method further includes transmitting repeatedly, by the beacon transmitter, a second advertising beacon signal through a second number of transmission repetitions at a second repeat interval, wherein the second advertising beacon signal includes an identifier that provides identification of the beacon transmitter and user information. The method further includes receiving a modification instruction from a user device. The modification instruction is configured to modify the first repeat interval into a third repeat interval. The method also includes transmitting repeatedly, by a beacon transmitter, the first beacon signal spaced at the third repeat interval.

[0004] In some embodiments, the first repeat interval is of a length different from a length of the second repeat interval.

[0005] In some embodiments, the first beacon signal is a signal alerting a receiving application on a personal wireless device to the presence of the beacon transmitter.

[0006] In some embodiments, the user information includes at least one selected from the group of user contact information and a user identifier.

[0007] In other embodiments, the first beacon signal is a signal using a first open protocol for alerting a receiving application to the presence of the beacon transmitter. The second advertising beacon signal is a signal using a second proprietary protocol for providing identification of the beacon transmitter.

[0008] In other embodiments, a location application function of the receiving application is activated in response to the first advertising beacon signal being received by the receiving application, and logging a location of the locator beacon by the location application function in response to the second beacon signal being received by the receiving application.

[0009] In other embodiment, the locator beacon is integrated within a power tool.

[0010] In other embodiments, the locator beacon is a standalone device configured to be mechanically coupled to an object.

[0011] In other embodiments, the locator beacon transmits using a wireless communication protocol.

[0012] In other embodiments, the wireless communication protocol is a Bluetooth communication protocol.

[0013] In some embodiments, a locator beacon is described. The locator beacon includes a wireless transmitter and one or more electronic processors. The electronic processors are configured to transmit repeatedly, via the wireless transmitter, a first advertising beacon signal through a first number of transmission repetitions spaced at a first repeat interval. The electronic processors are also configured to transmit repeatedly, via the wireless transmitter, a second advertising beacon signal through a second number of transmission repetitions at a second repeat interval, wherein the second advertising beacon signal includes an identifier that provides identification of the beacon transmitter and user information. The electronic processors are further configured to receive a modification instruction from a user device at the wireless transmitter, modify the first repeat interval into a third repeat interval based on the received modification instruction, and transmit repeatedly, by the wireless transmitter, the first beacon signal spaced at the third repeat interval.

[0014] In other embodiments, the first repeat interval is of a duration different from a duration of the second repeat interval.

[0015] In other embodiments, the user information includes at least one of user contact information and a user identifier.

[0016] In other embodiments, the first advertising beacon signal is a signal using a first open protocol for alerting a receiving application to the presence of the beacon transmitter, and the second advertising beacon signal is a signal using a second proprietary protocol for providing identification of the beacon transmitter.

[0017] In other embodiments, the locator beacon is integrated into a power tool.

[0018] In other embodiments, the wireless transmitter is a Bluetooth transmitter.

[0019] In some embodiments, a method for modifying a repeat interval of a locator beacon is described. The method includes transmitting repeatedly, by a wireless beacon transmitter, a first advertising beacon signal through a first number of transmission repetitions spaced at a first repeat interval. The method also includes transmitting repeatedly, by the wireless beacon transmitter, a second advertising beacon signal through a second number of transmission repetitions at a second repeat interval, wherein the second advertising beacon signal includes an identifier that provides identification of the beacon transmitter and user information. The method also includes receiving a modification instruction at the wireless beacon transmitter, the wireless beacon transmitter configured to modify the first repeat interval into a third repeat interval based on the received modification instruction. The method also includes, transmitting repeatedly, by the beacon transmitter, the first beacon signal spaced at the third repeat interval, wherein the third repeat interval is a longer duration than the first repeat interval.

[0020] In other embodiments, the method also includes receiving, at a wireless device, the first advertising beacon signal, activating, at the wireless device, a location application function in response to receiving the first advertising beacon signal, receiving at the wireless device, the second

advertising beacon signal, and logging, at the wireless device, a location of the locator beacon.

[0021] In other embodiments, the method also includes transmitting, by the wireless device, the logged location of the locator beacon to a location server.

[0022] In other embodiments, the location of the locator beacon is determined based on a location of the wireless device.

[0023] In other embodiments, the wireless beacon transmitter is a Bluetooth transmitter.

[0024] Before any embodiments are explained in detail, it is to be understood that the embodiments are not limited in its application to the details of the configuration and arrangement of components set forth in the following description or illustrated in the accompanying drawings. The embodiments are capable of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof are meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

[0025] In addition, it should be understood that embodiments may include hardware, software, and electronic components or modules that, for purposes of discussion, may be illustrated and described as if the majority of the components were implemented solely in hardware. However, one of ordinary skill in the art, and based on a reading of this detailed description, would recognize that, in at least one embodiment, the electronic-based aspects may be implemented in software (e.g., stored on non-transitory computer-readable medium) executable by one or more processing units, such as a microprocessor and/or application specific integrated circuits (“ASICs”). As such, it should be noted that a plurality of hardware and software based devices, as well as a plurality of different structural components, may be utilized to implement the embodiments. For example, “servers,” “computing devices,” “controllers,” “processors,” etc., described in the specification can include one or more processing units, one or more computer-readable medium modules, one or more input/output interfaces, and various connections (e.g., a system bus) connecting the components.

[0026] Relative terminology, such as, for example, “about,” “approximately,” “substantially,” etc., used in connection with a quantity or condition would be understood by those of ordinary skill to be inclusive of the stated value and has the meaning dictated by the context (e.g., the term includes at least the degree of error associated with the measurement accuracy, tolerances [e.g., manufacturing, assembly, use, etc.] associated with the particular value, etc.). Such terminology should also be considered as disclosing the range defined by the absolute values of the two endpoints. For example, the expression “from about 2 to about 4” also discloses the range “from 2 to 4”. The relative terminology may refer to plus or minus a percentage (e.g., 1%, 5%, 10%, or more) of an indicated value.

[0027] It should be understood that although certain drawings illustrate hardware and software located within particular devices, these depictions are for illustrative purposes only. Functionality described herein as being performed by

one component may be performed by multiple components in a distributed manner. Likewise, functionality performed by multiple components may be consolidated and performed by a single component. In some embodiments, the illustrated components may be combined or divided into separate software, firmware and/or hardware. For example, instead of being located within and performed by a single electronic processor, logic and processing may be distributed among multiple electronic processors. Regardless of how they are combined or divided, hardware and software components may be located on the same computing device or may be distributed among different computing devices connected by one or more networks or other suitable communication links. Similarly, a component described as performing particular functionality may also perform additional functionality not described herein. For example, a device or structure that is “configured” in a certain way is configured in at least that way but may also be configured in ways that are not explicitly listed.

[0028] Other aspects of the technology will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] FIG. 1A illustrates a diagram of a beacon transmitter for location reporting according to some embodiments.

[0030] FIG. 1B illustrates a front view of a beacon transmitter according to some embodiments.

[0031] FIG. 2 depicts an object location tracking system including the beacon transmitter of FIGS. 1A-1B attached to an object, according to some embodiments.

[0032] FIGS. 3A-3B illustrate an exemplary power tool incorporating the beacon transmitter of FIG. 1A for location reporting, according to some embodiments.

[0033] FIG. 4 illustrates an exemplary computer system for implementing location reporting, according to some embodiments.

[0034] FIG. 5 depicts a first beacon diagram for a first beacon signal transmittable by a beacon transmitter for location reporting, according to some embodiments.

[0035] FIG. 6 depicts a second beacon diagram for a second beacon signal transmittable by a beacon transmitter for location reporting, according to embodiments of the technology.

[0036] FIG. 7 is a flowchart for a method for logging a location of a beacon transmitter, according to some embodiments of the technology.

[0037] FIG. 8 is a flowchart for a method for modifying the transmission interval of a beacon transmitter, according to some embodiments.

[0038] FIGS. 9A, 9B, 9C, and 9D are timing diagrams for beacon transmissions, according to some embodiments.

DETAILED DESCRIPTION

[0039] FIG. 1A illustrates a beacon transmitter **100** for location reporting according to embodiments of the technology. The beacon transmitter **100**, also referred to as a transmitting device, includes a battery **110** (also referred to as a power source), a controller **125**, a power block **130**, a wireless transceiver **140**, an input/output (I/O) port **145**, a memory **160**, user input **155**, sensors **170**, and user output **175**. As described in further detail below, the beacon trans-

mitter repeatedly transmits, via the wireless transceiver **140**, a first beacon signal and a second advertising beacon signal according to a transmission pattern, which, in some embodiments, enables power-efficient location tracking of the beacon transmitter **100** and objects to which it is attached.

[0040] The battery **110** provides direct current (DC) power to the power block **130**. The battery **110** includes a housing within which is one or more battery cells, such as lithium ion (“Li-ion”) cells, Nickel-Cadmium (“Ni-Cad”) cells, or cells of another chemistry type. In some embodiments, the battery **110** is a coin cell battery. In some embodiments, the beacon transmitter **100** includes another power source in addition to the battery **110** or in place of the battery **110**, such as a circuit for connection to alternating current power (e.g., including a rectifier), photovoltaic cells and related circuitry for solar-based power generation, a wind-based power generator, or a kinetic energy power generator.

[0041] The power block **130** is coupled to the battery **110** via the terminals (not shown) of the battery **110** and matching terminals (not shown) of the power block **130**. The power block **130** provides DC power to components of the beacon transmitter **100**. The power block **130** may include power regulating and conversion circuitry to ensure that the power provided to various components of the beacon transmitter **100** is at the appropriate level(s).

[0042] The controller **125** is further coupled to the wireless transceiver **140** and the input/output (I/O) port **145**. As will be described in greater detail below, the power block **130**, wireless transceiver **140**, and I/O port **145** enable the beacon transmitter **100** to communicate with external devices and may be collectively referred to as a physical interface.

[0043] The controller **125**, which may be an electronic processor, is in communication with the memory **160**. In some embodiments, the memory **160** stores and provides to the controller **125** for transmission the data making up the first beacon signal and the second advertising beacon signal, which are described in further detail below. The memory **160** further includes, among other elements, instructions that are executed by the controller **125** to control the functions of the beacon transmitter **100** described herein. Although the instructions are described as software stored in memory **160** and executed by the controller **125**, the instructions may be implemented in part or wholly in hardware of the controller **125** or outside of the controller **125**. For example, the instructions may be implemented by one or more separate digital signal processors (DSPs) or general purpose processors executing the instructions, or by one or more field programmable gate arrays (FPGAs) or application specific integrated circuits (ASICs). Although the memory **160** is shown as a singular unit, the memory **160** may be made up of various memories individually coupled to the controller **125** or coupled to the controller **125** via a bus. Additionally, portions of the memory **160** may be embedded within the controller **125**. For instance, parameters such as a status of battery **110** may be stored within a memory of the controller **125**. The data stored in the memory **160** described herein may be provided from an external computing device via the wireless transceiver **140** or I/O port **145** and stored in the memory **160** by the controller **125**.

[0044] The user input **155** and sensors **170** include one or more of buttons, microphones, accelerometers, temperature sensors, humidity sensors, and light sensors capable of detecting external stimuli from the environment or the user.

The user output **175** includes one or more of LEDs, a speaker, a vibrating element, etc. to inform the user of the status of the beacon transmitter **100**. For example, when an error occurs, such as low battery power, the beacon transmitter **100** may output an audible alert, an LED may flash, and/or the vibrating element may provide tactile feedback to the user. The user output **175** may be controlled by output signals from the controller **125**.

[0045] The controller **125** is further coupled to the wireless transceiver **140** and the I/O port **145**. As described in further detail below, the controller **125** may transmit wireless communications via the wireless transceiver **140** and may receive wireless communications via the wireless transceiver **140**. The I/O port **145** may include a wired connection for the beacon transmitter **100** to enable, for example, programming of the beacon transmitter **100** or data export from the beacon transmitter **100**. In some embodiments, the wireless transceiver **140** is configured to transmit and receive the wireless communications in accordance with Bluetooth and/or Bluetooth low energy (“BLE”) protocols, or other short-range wireless protocols. In some embodiments, additionally or alternatively, the wireless transceiver **140** is configured to communication via cellular communications (e.g., 3G, 4G, 5G, LTE, CDMA, etc.), or other applicable communication protocols. In some embodiments, the wireless transceiver **140** is also configured to receive positional data, such as satellite positional data (e.g., GPS). In some embodiments, the cellular and/or positional data may only be available where the beacon transmitter **100** is integrated into another device, such as a power tool as described below, where there is additional power available, such as via the power tool battery.

[0046] FIG. 1B illustrates an embodiment of the beacon transmitter **100** including a housing **180** with mounting holes **182**. The various components of the beacon transmitter **100** illustrated in FIG. 1A are located within and supported by the housing **180**. The mounting holes **182** are configured to receive fasteners (e.g., screws) to secure the beacon transmitter **100** to an object to be tracked. Other securing elements are used in some embodiments, such as an adhesive pad on the back of the housing **180** (not shown). In some embodiments, the beacon transmitter **100** includes a housing having one or more of a different shape, differently positioned mounting holes, and different elements for mounting to objects.

[0047] FIG. 2 depicts an object location tracking system **200** including the beacon transmitter **100** attached to an object **210**, illustrated as a ladder. In some embodiments, the beacon transmitter **100** is secured to the object **210** using an adhesive, hook and loop fasteners, or the like, rather than via fasteners through the mounting holes **182**. The beacon transmitter **100** communicates via wireless signals **202** (e.g., Bluetooth™ low energy transmissions) with a personal wireless device **204** configured to receive such signals. The personal wireless device **204** (also referred to as a receiving device) may be, for example, a mobile smart phone, laptop computer, desktop computer, personal digital assistant (PDA), or other receiving device. Personal wireless device **204** communicates via a network **206** with a location server **208**. Example computer systems that may implement personal wireless device and location server **208** are discussed below with respect to FIG. 4. The network **206** may include

one more of a local area network (LAN), wide area network (WAN) (e.g., the Internet), a cellular network, or other networks.

[0048] In some embodiments, the beacon transmitter **100** is integrated within an object to be tracked. For example, with respect to FIGS. 3A-B, the beacon transmitter **100** is integrated into a power tool **300**, which is described in further detail below.

[0049] As described in further detail below, the personal wireless device **204** receives beacon data from the beacon transmitter **100** via the wireless signals **202**. The beacon data may include one or more of a transmitter identifier, a user identifier, user contact information, timestamp, state of charge of the battery **110**, an object identifier (identifying the object **210**), and other status information. In turn, the personal wireless device **204** (a) logs the beacon data locally on a memory of the personal wireless device **204**, (b) sends tracking data, based on the beacon data, to the location server **208** for logging, or (c) both logs the beacon data and sends the tracking data.

[0050] The location server **208** includes a tracking database **212**. A tracking application may be executed by a processor of the location server **208** to receive tracking data from the personal wireless device **204**, update the tracking database **212**, and to receive and respond to database queries for the tracking database **212**. The tracking database **212** stores tracking data for the beacon transmitter **100** including one or more of a transmitter identifier, a user identifier (e.g., an owner of the beacon transmitter **100**), user contact information, timestamp, last known location, state of charge of the battery **110**, other status information, personal wireless device identifier (e.g., identifying the most recent personal wireless device **204** that received communications from the beacon transmitter and communicated to the location server **208**), and location history (e.g., including previous known locations, timestamps, and personal wireless device identifiers). The tracking database **212** also stores a lost/not-lost indication (e.g., a flag) that indicates, based on a value of the indicator, whether the beacon transmitter **100** is considered “lost” or “not lost.”

[0051] Although a single beacon transmitter **100** is illustrated in FIG. 2, in some embodiments, the system **200** includes a plurality of beacon transmitters **100**, each used to track a different object. Similarly, although a single personal wireless device **204** is illustrated in FIG. 2, in some embodiments, the system **200** includes a plurality of personal wireless devices **204** that may each receive wireless signals **202** from one or more of the personal wireless devices **204** and that may each communicate with the location server **208** over the network **206** or another network. Accordingly, the tracking database **212** stores and updates tracking data for each beacon transmitter **100** in the system **200** based on communications from the one or more personal wireless devices **204**.

[0052] Although the location server **208** is illustrated as a singular unit, the location server **208** may be made up of various servers located together or remotely and coupled via one or more networks. Similarly, the tracking database **212** may be a single database or made up of various databases in communication with one another.

[0053] Although the object **210** is illustrated in FIG. 2 as being a ladder, the beacon transmitter **100** may be mounted on various other objects including other types of tools and accessories. For example, the beacon transmitter **100** may be

mounted on and used with hand tools, power tools, test and measurement equipment, battery packs, vacuum cleaners, work site radios, outdoor power equipment, and vehicles. Other tools on which versions of the beacon transmitter **100** may be mounted include drills, circular saws, jig saws, band saws, reciprocating saws, screw drivers, angle grinders, straight grinders, hammers, multi-tools, impact wrenches, rotary hammers, impact drivers, angle drills, pipe cutters, grease guns, and the like. Test and measurement equipment can include digital multimeters, clamp meters, fork meters, wall scanners, IR thermometers, laser distance meters, laser levels, remote displays, insulation testers, moisture meters, thermal imagers, inspection cameras, and the like. Vacuum cleaners can include stick vacuums, hand vacuums, upright vacuums, carpet cleaners, hard surface cleaners, canister vacuums, broom vacuums, and the like. Outdoor power equipment can include blowers, chain saws, edgers, hedge trimmers, lawn mowers, trimmers, and the like. Other tools on which versions of the transmitter may be mounted include devices such as electronic key boxes, calculators, cellular phones, head phones, cameras, motion sensing alarms, flashlights, worklights, weather information display devices, a portable power source, a digital camera, a digital music player, a radio, and multi-purpose cutters.

[0054] FIGS. 3A-B illustrate a power tool **300** incorporating the beacon transmitter **100** described above. The power tool **300** includes a power tool battery pack or other power tool power supply **302**, a tool controller **304**, a tool motor **306**, and a tool output component **308**. The power tool power supply **302** may include a circuit for connection to alternating current power, may include power generation components, such as a wind or solar generator, or may be a battery pack that may include one or more battery cells (e.g., lithium-ion cells) within a housing that includes contacts and an attachment mechanism for selectively securing and removing the power tool battery pack to the power tool **300**. The tool controller **304** is coupled to and powered by the power tool power supply **302**, and controls the tool motor **306** to drive the tool output component **308**. The tool output component **308** may be, for example, a drill chuck, as illustrated in FIG. 3B. The tool controller **304** may control the tool motor **306** based on user input received via user input component **310**, which may be, for example, a trigger, as illustrated in FIG. 3B. The power tool **300** may further include a tool housing **312** (FIG. 3B that houses the tool controller **304**, the tool motor **306**, and the beacon transmitter **100**).

[0055] The beacon transmitter **100** may be coupled to the tool controller **304** and the power tool power supply **302**. For example, the beacon transmitter **100** may be powered by the power tool power supply **302** when present, and by the battery **110** of the beacon transmitter **100** when the power tool power supply **302** is not coupled to the power tool **300**. Additionally, the beacon transmitter **100** may communicate with the tool controller **304** to, for example, (i) obtain tool usage data stored on a memory of the tool controller **304** (e.g., obtained by sensors of the power tool **300**) to send to the personal wireless device **204** and/or (ii) provide tool configuration data (e.g., that is sent to the tool controller **304** for storage on a memory thereof) received from the personal wireless device **204**. The beacon transmitter **100**, when incorporated into the power tool **300**, may store within the memory **160** (see FIG. 1) identifying information for the power tool **300**, such as a product identifier (e.g., identifying

the type of power tool) and a serial number (e.g., uniquely identifying the particular instance of the power tool). This identifying information of the power tool **300** may also be provided by the beacon transmitter **100** as part of the beacon data transmitted with the second advertising beacon signal, described below.

[0056] The power tool **300**, as illustrated in FIG. 3B, is a hammer drill/driver. However, the power tool **300** is merely an example, and other power tools may have the beacon transmitter **100** incorporated therein. Additionally, other devices may have the beacon transmitter **100** incorporated therein, such as test and measurement equipment, battery packs (e.g., the power tool power supply **302**), vacuum cleaners, work site radios, work site lights, outdoor power equipment, and vehicles. Such an incorporated beacon transmitter **100** may be powered by a battery of the device in which the beacon transmitter **100** is incorporated, similar to that which is described with respect to the power tool **300**.

[0057] Further, in some embodiments, the beacon transmitter **100** is incorporated into a repeater device that receives other beacon signals (e.g., similar to the beacon signals emitted by the beacon transmitter **100**) and repeats (i.e., transmits) those beacon signals using beaconing techniques as described herein.

[0058] Embodiments of location logging module and/or of the various location logging methods and techniques as described herein may be executed on one or more computer systems, which may interact with various other devices. One such computer system **400** is illustrated in FIG. 4. In different embodiments, computer system **400** may be any of various types of devices, including, but not limited to, a personal computer system, desktop computer, laptop, notebook, or netbook computer, mainframe computer system, handheld computer, mobile telephone, workstation, network computer, a camera, a set top box, a mobile device, a consumer device, video game console, handheld video game device, application server, storage device, a peripheral device such as a switch, modem, router, or another type of computing or electronic device. The computer system **400** is an example of a computer system that may be configured to implement the location server **208**, and of a computer system that may be configured to implement the personal wireless device **204**.

[0059] In the illustrated embodiment, the computer system **400** includes one or more processors **410** coupled to a system memory **420** via an input/output (I/O) interface **422**. Computer system **400** further includes a network interface **428** coupled to I/O interface **422**, and one or more input/output devices **430**, such as cursor control device **432**, keyboard **434**, and display(s) **436**. In some embodiments, it is contemplated that embodiments may be implemented using a single instance of computer system **400**, while in other embodiments multiple such systems, or multiple nodes making up computer system **400**, may be configured to host different portions or instances of embodiments. For example, in one embodiment some elements may be implemented via one or more nodes of computer system **400** that are distinct from those nodes implementing other elements.

[0060] In various embodiments, computer system **400** may be a uniprocessor system including one processor **410**, or a multiprocessor system including several processors **410** (e.g., two, four, eight, or another suitable number). Processors **410** may be any suitable processor capable of executing instructions. For example, in various embodiments, proces-

sors **410** may be general-purpose or embedded processors implementing any of a variety of instruction set architectures (ISAs), such as the x86, PowerPC, SPARC, or MIPS ISAs, or any other suitable ISA. In multiprocessor systems, each of processors **410** may commonly, but not necessarily, implement the same ISA.

[0061] In some embodiments, at least one processor **410** may be a graphics processing unit. A graphics processing unit or GPU may be considered a dedicated graphics-rendering device for a personal computer, workstation, game console or other computing or electronic device. Modern GPUs may be very efficient at manipulating and displaying computer graphics, and their highly parallel structure may make them more effective than typical CPUs for a range of complex graphical algorithms. For example, a graphics processor may implement a number of graphics primitive operations in a way that makes executing them much faster than drawing directly to the screen with a host central processing unit (CPU). In various embodiments, the image processing methods disclosed herein may, at least in part, be implemented by program instructions configured for execution on one of, or parallel execution on two or more of, such GPUs. The GPU(s) may implement one or more application programmer interfaces (APIs) that permit programmers to invoke the functionality of the GPU(s). Suitable GPUs may be commercially available from vendors such as NVIDIA Corporation, ATI Technologies (AMD), and others.

[0062] System memory **420** may be configured to store program instructions and/or data accessible by processor **410**. In various embodiments, system memory **420** may be implemented using any suitable memory technology, such as static random access memory (SRAM), synchronous dynamic RAM (SDRAM), nonvolatile/Flash-type memory, or any other type of memory. In the illustrated embodiment, program instructions and data implementing desired functions, such as those described above for various embodiments, are shown stored within system memory **420** as program instructions **424** and data storage **426**, respectively. In other embodiments, program instructions and/or data may be received, sent or stored upon different types of computer-accessible media or on similar media separate from system memory **420** or computer system **400**. Generally speaking, a computer-accessible medium may include storage media or memory media such as magnetic or optical media, e.g., disk or CD/DVD-ROM coupled to computer system **400** via I/O interface **422**. Program instructions and data stored via a computer-accessible medium may be transmitted by transmission media or signals such as electrical, electromagnetic, or digital signals, which may be conveyed via a communication medium such as a network and/or a wireless link, such as may be implemented via network interface **428**.

[0063] In one embodiment, I/O interface **422** may be configured to coordinate I/O traffic between processor **410**, system memory **420**, and any peripheral devices in the device, including network interface **428** or other peripheral interfaces, such as input/output devices **430**. In some embodiments, I/O interface **422** may perform any necessary protocol, timing or other data transformations to convert data signals from one component (e.g., system memory **420**) into a format suitable for use by another component (e.g., processor **410**). In some embodiments, I/O interface **422** may include support for devices attached through various types of peripheral buses, such as a variant of the Peripheral Component Interconnect (PCI) bus standard or the Universal

Serial Bus (USB) standard, for example. In some embodiments, the function of I/O interface 422 may be split into two or more separate components, such as a north bridge and a south bridge, for example. In addition, in some embodiments some or all of the functionality of I/O interface 422, such as an interface to system memory 420, may be incorporated directly into processor 410.

[0064] Network interface 428 may be configured to allow data to be exchanged between computer system 400 and other devices attached to a network, such as other computer systems, or between nodes of computer system 400. In various embodiments, network interface 428 may support communication via wired or wireless general data networks, such as any suitable type of Ethernet network, for example; via telecommunications/telephony networks such as analog voice networks or digital fiber communications networks; via storage area networks such as Fibre Channel SANs, or via any other suitable type of network and/or protocol.

[0065] For example, when the computer system 400 implements the personal wireless device 204, the network interface 428 may include one or more wireless antennas to enable wireless communication with the beacon transmitter 100 and the location server 208. Additionally, when the computer system 400 implements the location server 208, the network interface 428 may include one or more wireless antennas to enable wireless communication with the personal wireless device 204.

[0066] Input/output devices 430 may, in some embodiments, include one or more display terminals, keyboards, keypads, touchpads, scanning devices, voice or optical recognition devices, or any other devices suitable for entering or retrieving data by one or more computer system 400. Multiple input/output devices 430 may be present in computer system 400 or may be distributed on various nodes of computer system 400. In some embodiments, similar input/output devices may be separate from computer system 400 and may interact with one or more nodes of computer system 400 through a wired or wireless connection, such as over network interface 428.

[0067] As shown in FIG. 4, the computer system 400 may further include a global navigation satellite system (GNSS) receiver 438. The GNSS receiver 438 is configured to receive signals from global navigation satellites and to determine, based on the received signals, a location of the GNSS receiver 438 (e.g., including a latitude, longitude, and altitude) and time. The GNSS receiver 438 is further configured to provide the determined location and time to other components of the computer system 400, such as the processor(s) 410. When the computer system 400 implements the personal wireless device 204, the determined location and time information may be used as the location and time of the personal wireless device 204 used in the various embodiments described herein. The GNSS receiver may, in some embodiments, be a global positions system (GPS) receiver.

[0068] FIG. 5 depicts a first beacon diagram 500, which represents example contents of the first beacon signals transmitted by the beacon transmitter 100 according to embodiments of the technology. The example first beacon diagram 500 illustrates the iBeacon™ protocol and includes a 47-byte transmission including a 1-byte preamble 502, a 4-byte access address 504, which is generally set to the value 0x8E89BED6, a protocol data unit (PDU) 506 of 2-39 bytes, and a cyclic redundancy check (CRC) 508 of 3 bytes.

[0069] PDU 506 includes a header 510 of 2 bytes, a MAC address 512 of 6 bytes, and data 514 of 0-31 bytes. Data 514 includes an iBeacon™ prefix 516 of 9 bytes, a universally unique identifier (UUID) 518 of 16 bytes, a major component 520 of 2 bytes, a minor component 522 of 2 bytes, and a transmission power component 524 of 1 bytes. The UUID 518 may identify, uniquely, the device transmitting the signal (e.g., the beacon transmitter 100). In some embodiments, the first beacon signals may take the form of another open protocol different than that which is illustrated in FIG. 5. For example, the particular fields and the lengths of the fields (e.g., the number of bytes) in the first beacon signals illustrated in FIG. 5 is an example, and some embodiments include additional fields, fewer fields, alternate fields, or fields with different lengths.

[0070] FIG. 6 depicts a second beacon diagram 600, which represents example contents of the second advertising beacon signals transmitted by the beacon transmitter 100 according to embodiments of the technology. The second beacon diagram 600 has various segments including a uniquely identifying product identifier (ID) 602 that uniquely identifies the type of device transmitting the signal (e.g., the model of the beacon transmitter 100), a serial number 604 that uniquely identifies the particular device from other devices of a similar type, and a universally unique identifier (UUID) 606 that uniquely identifies the particular device transmitting the signal (e.g., the beacon transmitter 100). The second beacon diagram 600 also includes other segments 608 as well, which may include data representing one or more of a user identifier, user contact information, timestamp, state of charge of the battery 110, and other status information. In some embodiments, the second advertising beacon signals may take the form of another proprietary protocol different than that which is illustrated in FIG. 6. For example, the particular fields and the lengths of the fields (e.g., the number of bytes) in the second advertising beacon signals illustrated in FIG. 6 is an example, and some embodiments include additional fields, fewer fields, alternate fields, or fields with different lengths.

[0071] FIG. 7 is a flowchart for a method for implementing location reporting for a receiving device in communication with a beacon transmitter according to some embodiments. The method of FIG. 7 is described with respect to the system 200; however, the method may be similarly applicable to other devices and systems. At process block 700, the personal wireless device 204 monitors for a beacon signal. In some embodiments, the personal wireless device 204 may monitor for a beacon signal by simply passively monitoring for one or more beacon signals via the network interface 428. As described above, the beacon signal may be transmitted using various communication protocols, which may be monitored via the personal wireless device. At process block 702, the personal wireless device determines whether a first beacon signal has been received, such as one of the first beacon signals 500. In response to determining that the first beacon signal was not received, the personal wireless device then determines whether a dormancy timer has elapsed. The dormancy timer may be a time period since the personal wireless device last received a first beacon signal. In some embodiments, the dormancy timer has a predetermined value, such as ten seconds. However, time periods of more than ten seconds or less than ten seconds are also contemplated. In response to determining that the dormancy timer has not elapsed, the personal wireless device 204

continues to monitor for a beacon signal at process block 700. In response to determining that the dormancy timer has elapsed, the personal wireless device 204 returns a transmitter location logging application to a dormant state. The personal wireless device 204 then continues to monitor for a beacon signal at process block 703.

[0072] Responsive to determining that a first beacon signal has been received at process block 701, the personal wireless device 204 determines whether a transmitter location logging application of the personal wireless device 204 is active at process block 704. In some embodiments, responsive to determining that the first beacon signal has been received at process block 701, the personal wireless device 204 also resets the dormancy timer. When the transmitter location logging application is determined to be active, the personal wireless device 204 returns to monitoring for a beacon signal at process block 700.

[0073] In response to the personal wireless device 204 determining that the location application is not active, the location logging application is activated at process block 706. In some embodiment, an operating system of the personal wireless device 204 receives the first beacon signal and activates the dormant transmitter location logging application. In other embodiments, the first beacon is a signal using a first open protocol for alerting the transmitter location logging application to the presence of the beacon transmitter 100 transmitting the second advertising beacon signal. In one embodiment, the first beacon signal is structured similarly to the first beacon diagram described above. In one embodiment, the location logging application is stored in a memory of the personal wireless device 204, and is configured to store a location of the personal wireless device 204 when the personal wireless device 204 receiver is activated at process block 706.

[0074] The transmitter location logging application of the personal wireless device 204 listens for a second advertising beacon signal at process block 708. For example, to listen, the personal wireless device 204 may execute a software loop that repeatedly checks for a second advertising beacon signal being received by a wireless antenna of the personal wireless device 204. In block 710, the personal wireless device 204 receives the second advertising beacon signal, such as one of the second beacon signals described above, from the transmitting device.

[0075] In block 712, a location of the beacon transmitter 100 that transmitted the second advertising beacon signal is logged by the transmitter location logging application of the personal wireless device 204. For example, upon receipt of one of the second advertising beacon signals including the beacon data, the personal wireless device 204 determines the transmitter identifier of the beacon transmitter 100 based on beacon data, and determines the location of the personal wireless device 204 based on an output from a global navigation satellite system (GNSS) receiver of the personal wireless device 204. While GNSS is described as an example of location detection, embodiments will include other forms of location awareness, such as registration of location (e.g., stored in a memory as part of an initial set-up) or location detection through detection of wireless networks, without departing from the scope and intent of the present disclosure. The personal wireless device 204 logs (e.g., stores in a memory) the determined location with the transmitter identifier of the beacon transmitter 100 such that the

location of the personal wireless device 204 is logged as the location of the beacon transmitter.

[0076] In some embodiments, the additional information may be logged by the personal wireless device 204 for the beacon transmitter 100 at process block 712. For example, additional information from the beacon data, including one or more of a user identifier, user contact information, timestamp, state of charge of the battery 110, and other status information may be logged for the beacon transmitter 100 at process block 712. Furthermore, additional information from the personal wireless device 204 may be logged for the beacon transmitter 100 at process block 712, such as a timestamp (e.g., when not provided as part of the beacon data) and receiving a device identifier that identifies the personal wireless device 204 or the user thereof. The data that is logged by the personal wireless device 204 in block 712 may be referred to as logged data from the beacon transmitter 100.

[0077] In some embodiments, the personal wireless device 204 further sends the logged data, including the transmitter identifier and location of the beacon transmitter 100 to the location server 208 for storage and processing. In some embodiments, the receiving device sends the logged data to the location server 208 each time process block 712 is executed. In other embodiments, the personal wireless device 204 may be configured to delay sending the logged data when the receiving device has already sent similar data recently (e.g., within the past minute, ten minutes, or hour) to limit data transmission and conserve power. In some embodiments, delaying transmission of the logged data enables the receiving device to obtain further logged data from other beacon transmitters using a similar process and to bundle the logged data for multiple beacon transmitters for a single transmission.

[0078] In some embodiments, the process described in FIG. 7 further includes, during execution of steps 700-712, the personal wireless device 204 waiting through a first number of transmission repetitions of the beacon transmitter 100 spaced at a first repeat interval, after the first number or transmission repetitions, waiting during a transition interval, and, thereafter, receiving the second advertising beacon signal through a second number of transmission repetitions of the beacon transmitter 100 spaced at a second repeat interval.

[0079] In some embodiments, the method of FIG. 7 occurs in the background of the personal wireless device 204 such that the receiving and logging of information related to the beacon transmitter 100 occurs without particular notifications to a user of the personal wireless device 204 of the particular receiving and logging. For example, although the transmitter location logging application may be activated from a dormant state, the activation may occur in the background such that an application on the personal wireless device 204 is not interrupted or altered to provide a notification of the activation. Similarly, the logged data may be logged on the personal wireless device 204 and sent to the location server 208 for logging without a particular notification of these actions being provided to the user of the personal wireless device 204.

[0080] In some embodiments, in addition to being able to be activated upon receipt of the first beacon signal, the transmitter location logging application of the personal wireless device 204 may also be activated in response to receiving a user activation input through a user interface. For

example, the user activation input may include a user input indicating a selection of the transmitter location logging application for execution. In response to the user activation input, the receiving device proceeds to block **708-712**, as described above.

[0081] The above process is useful for determining the last location of a beacon transmitter **100**. However, as shown in the above process, when the transmitter location logging application is already active and a first beacon signal is received in block **701**, the beacon transmitter returns to block **700** to monitor for a further beacon signal, rather than proceed to blocks **706-712** to log a location of the beacon transmitter that transmitted the signal. Accordingly, after the application is activated and the location of a first beacon transmitter is logged, the location of another beacon transmitter is not logged until the application returns to a dormant state (after the dormancy timer elapses because a beacon signal was not received for a predetermined period of time (the length of the dormancy timer). For example, after receiving a beacon signal, the transmitter location logging application may go dormant after not receiving a further beacon signal for one second. However, predetermined times of more than one second or less than one second are also contemplated. In some embodiments, the timing of the beacon signal transmitted by the beacon transmitter may be such that the interval of transmission is generally longer than the predetermined time period for the transmitter location logging application to go dormant. However, when there are multiple beacon transmitter collocated near the user device **204**, the overlapping transmissions may not allow the user device **204** to stop receiving the first beacon signals (from one or more beacon transmitters **100**) long enough for the transmitter location logging application to go dormant. Accordingly, the beacon signals from other beacon transmitters **100** may not result in logging the location of those other beacon transmitters **100** because block **712** of FIG. **7** is not reached.

[0082] To address this issue, a process **800** is described below in regards to FIG. **8**, which allows for one or more parameters of the beacon signal to be modified. At process block **802**, a beacon transmitter, such as beacon transmitter **100** described above transmits a beacon signal at a default interval. The beacon signal may have a unique beacon ID. For example, the beacon signal may have a unique iBeacon™ prefix or a universally unique identifier (UUID), such as those described in FIG. **5**, above. In one embodiment, the beacon signal may be transmitted once every **15** seconds by the beacon transmitter **100**. For example, FIG. **9A** illustrates a timing diagram for a series of transmissions **900** output by a beacon transmitter. The transmissions may be the first beacon signal, such as that described in FIG. **5** above. Each transmission **900** has the same unique beacon ID. In one embodiment, the timing diagram of FIG. **9A** illustrates a default transmission interval used by beacon transmitters. At process block **804**, the beacon transmitter **100** receives a modification instruction. In one embodiment, the modification instruction may be received from a user device, such as user device **204**. For example, the transmitter location logging application may be configured to allow a user to manually modify a beacon signal output by the beacon transmitter **100**. In one embodiment, the transmitter location logging application is configured to allow a user to modify the transmission interval of a beacon signal transmitted by a beacon transmitter. In further embodiments, the transmitter

location logging application is also configured to modify the unique beacon ID of a beacon signal. However, in other embodiments, the unique beacon ID associated with a beacon transmitter **100** is hardcoded, and cannot be changed.

[0083] At process block **806**, the beacon transmitter **100** determines whether the modification instructions modified any of the parameters (e.g. timing interval, unique ID, etc.) to values other than the default or current parameter settings. Based on determining that there is no difference between the current parameters of the beacon transmitter **100** and the received modification instructions, the beacon transmitter **100** continues to transmit the beacon signal at process block **802**. When the beacon transmitter **100** determines that the beacon parameters have been modified to values different than the default or current parameter settings, the beacon transmitter **100** modifies the beacon parameters at process block **808**. As described above, the modification instructions include instructions to modify a transmission interval. Once the beacon transmitter **100** has modified the parameters, the beacon transmitter **100** transmits the beacon signal using the modified parameters at process block **810**. While the parameters, such as the timing interval of the first beacon signal are shown as able to be modified using the process **800**, the second advertising beacon signal may be configured to remain the same, regardless of the modified first beacon signal. In other embodiments, the second advertising beacon signal dynamically changes based on the transmission interval of the first beacon signal. For example, the second advertising beacon signal may be determined as a fraction of the transmission interval of the first beacon signal.

[0084] For example, the second advertising beacon may have a transmission interval $\frac{1}{50}^{th}$ the frequency of the transmission interval of the first beacon signal. However, various other ratios of greater than $\frac{1}{50}^{th}$ and less than $\frac{1}{50}^{th}$ of the transmission interval of the first beacon signal are also contemplated.

[0085] Turning now to FIGS. **9B** and **9C**, modified beacon transmission intervals are shown, according to some embodiments. In FIG. **9B** the transmission interval of a beacon signal **902** has been modified to be 60 seconds. Further, FIG. **9C** illustrates a beacon signal **904** with a transmission interval modified to 300 seconds. The transmission intervals shown in FIGS. **9A-9C** may each represent different beacon transmitters, or a single beacon transmitter modified in different ways based on different usage scenarios. For example, the timing interval of FIG. **9A** may be useful when the personal wireless device **204** is only in the vicinity of the beacon transmitter **100** for a short period of time, such as when the user associated with the personal wireless device **204** may only be near the beacon transmitter **100** for short durations of time, such as when walking into or out of a jobsite or tool crib. The timing interval of FIG. **9B** may be useful when the personal wireless device **204** is in a generally static environment (e.g. generally around the beacon transmitter), where there is less risk of the user walking by or moving out of range of the beacon transmitter without receiving the beacon signal. The timing interval of FIG. **9B** is also advantageous where the personal wireless device is in a generally static environment and there are a limited number of other beacon transmitters in the area.

[0086] Finally, the timing interval of FIG. **9C** may be useful when the personal wireless device **204** is in a static environment with many other beacon transmitters. By expanding the timing interval as shown in FIG. **9C** for

multiple beacon transmitters in close proximity, the transmitter location logging application of the personal wireless device **204** will generally have enough time between received beacon signals to go dormant, allowing for it to subsequently be woken up by a received beacon signal. The longer interval of FIG. 9C allows for a higher likelihood that multiple beacon transmitters **100** have their location logged by the transmitter location logging application when there are multiple beacon transmitters **100** in close proximity to each other.

[0087] Turning now to FIG. 9D, a combination timing interval **906** is shown combining the timing intervals of FIGS. 9A-9C. The combination timing interval **906** provides an example of what would be received by the personal wireless device **204** when there are multiple beacon transmitters with multiple timing intervals. In the example of FIG. 9D, each of the beacon signals **900**, **902**, **904** may be received by the personal wireless device **204**. However, due to the staggered timing intervals the likelihood that each individual beacon signal **900**, **902**, and **904** will be received by the personal wireless device **204** when the transmitter location logging application is dormant is increased. Further, the beacon signals **900**, **902**, **904** may each have a unique beacon ID. However, in some embodiments, the same beacon ID may be used for one or more of the beacon signals **900**, **902**, **904**.

[0088] Those skilled in the art will also appreciate that, while various items are illustrated as being stored in memory or on storage while being used, these items or portions of them may be transferred between memory and other storage devices for purposes of memory management and data integrity. Alternatively, in other embodiments some or all of the software components may execute in memory on another device and communicate with the illustrated computer system via inter-computer communication. Some or all of the system components or data structures may also be stored (e.g., as instructions or structured data) on a computer-accessible medium or a portable article to be read by an appropriate drive, various examples of which are described above. In some embodiments, instructions stored on a computer-accessible medium separate from computer system **400** may be transmitted to computer system **400** via transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link. Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Accordingly, the present technology may be practiced with other computer system configurations.

[0089] Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Generally speaking, a computer-accessible medium may include storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD-ROM, volatile or non-volatile media such as RAM (e.g. SDRAM, DDR, RDRAM, SRAM, etc.), ROM, etc., as well as transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as network and/or a wireless link.

[0090] The various methods as illustrated in the figures and described herein represent example embodiments of

methods. The methods may be implemented in software, hardware, or a combination thereof. The order of method may be changed, and various elements may be added, reordered, combined, omitted, modified, etc.

[0091] Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. It is intended that the technology embrace all such modifications and changes and, accordingly, the above description to be regarded in an illustrative rather than a restrictive sense.

[0092] Thus, the technology provides, among other things, systems and methods for location logging of transmission devices. Various features and advantages of the technology are set forth in the following claims.

1.-20. (canceled)

21. A tracking system comprising:

a locator beacon including:

a first wireless transmitter; and

one or more first electronic processors, the one or more first electronic processors configured to:

transmit repeatedly, via the first wireless transmitter, a first beacon signal through a first number of transmission repetitions spaced at a first repeat interval,

transmit repeatedly, via the first wireless transmitter, a first advertising signal through a second number of transmission repetitions at a second repeat interval, wherein the first advertising signal includes an identifier that provides identification of the locator beacon, and

transmit repeatedly, via the first wireless transmitter, a second beacon signal through a third number of transmission repetitions spaced at a third repeat interval, wherein the third repeat interval is different than the first repeat interval, and wherein at least one of the first number of transmission repetitions occurs between at least two of the third number of transmission repetitions; and

a wireless device including:

a second wireless transmitter, and

one or more second electronic processors, the one or more second electronic processors configured to:

activate, in response to either one of the first beacon signal and the second beacon signal being received by the wireless device, a location logging application, and

log a location of the locator beacon by the location logging application in response to the first advertising signal being received by the wireless device while the location logging application is active.

22. The tracking system of claim 21, wherein the one or more first electronic processors is configured to:

transmit repeatedly, via the first wireless transmitter, a third beacon signal through a fifth number of transmission repetitions spaced at a fifth repeat interval, wherein the fifth repeat interval is different than the first repeat interval and the third repeat interval, and wherein at least one of the third number of transmission repetitions occurs between at least two of the fifth number of transmission repetitions;

wherein the one or more second electronic processors is configured to activate, in response to the third beacon signal being received by the wireless device, the location logging application.

23. The tracking system of claim 21, wherein the one or more first electronic processors is configured to:

receive, via the first wireless transmitter, a modification instruction from the wireless device or another wireless device; and

modify the first repeat interval, the third repeat interval, or both the first repeat interval and the third repeat interval based on the modification instruction.

24. The tracking system of claim 21, wherein the first beacon signal includes a first beacon identification, and wherein the second beacon signal includes a second beacon identification different than the first beacon identification.

25. The tracking system of claim 21, wherein the second repeat interval is based on the first repeat interval.

26. The tracking system of claim 21, wherein:

the first beacon signal and the second beacon signal are each a first signal using an open protocol for alerting the location logging application to a presence of the locator beacon; and

the first advertising signal is a second signal using a proprietary protocol for providing the identification of the locator beacon.

27. The tracking system of claim 21, wherein the location of the locator beacon is determined based on a location of the wireless device.

28. The tracking system of claim 21, wherein the locator beacon is integrated within a power tool.

29. The tracking system of claim 21, wherein a second location of the locator beacon or another locator beacon is logged by the location logging application in response to the wireless device receiving another first advertising beacon signal from the locator beacon or a second advertising signal from the another locator beacon, and

wherein the second location of the locator beacon or the another locator beacon is not logged until (i) the location logging application returns to a dormant state after being active, (ii) the wireless device receives, when the location logging application is in the dormant state, another of either one of the first beacon signal and the second beacon signal from the locator beacon or another beacon signal from the another locator beacon, and (iii) the wireless device re-activates the location logging application in response to receiving the another of either one of the first beacon signal and the second beacon signal or the another beacon signal.

30. A method of tracking a locator beacon, the method comprising:

transmitting repeatedly, with one or more first electronic processors of the locator beacon via a first wireless transmitter of the locator beacon, a first beacon signal through a first number of transmission repetitions spaced at a first repeat interval;

transmitting repeatedly, with the one or more first electronic processors via the first wireless transmitter, a first advertising signal through a second number of transmission repetitions at a second repeat interval, wherein the first advertising signal includes an identifier that provides identification of the locator beacon;

transmitting repeatedly, with the one or more first electronic processors via the first wireless transmitter, a second beacon signal through a third number of transmission repetitions spaced at a third repeat interval, wherein the third repeat interval is different than the first repeat interval, and wherein at least one of the first

number of transmission repetitions occurs between at least two of the third number of transmission repetitions;

activating, with one or more second electronic processors of a wireless device and in response to either one of the first beacon signal and the second beacon signal being received by the wireless device via a second wireless transmitter of the wireless device, a location logging application; and

logging, with the one or more second electronic processors, a location of the locator beacon by the location logging application in response to the first advertising signal being received by the wireless device via the second wireless transmitter while the location logging application is active.

31. The method of claim 30, further comprising:

transmitting repeatedly, with the one or more first electronic processors via the first wireless transmitter, a third beacon signal through a fifth number of transmission repetitions spaced at a fifth repeat interval, wherein the fifth repeat interval is different than the first repeat interval and the third repeat interval, and wherein at least one of the third number of transmission repetitions occurs between at least two of the fifth number of transmission repetitions; and

activating, with the one or more second electronic processors and in response to the third beacon signal being received by the wireless device via the second wireless transmitter, the location logging application.

32. The method of claim 30, further comprising:

receiving, with the one or more first electronic processors via the first wireless transmitter, a modification instruction from the wireless device or another wireless device; and

modifying, with the one or more first electronic processors, the first repeat interval, the third repeat interval, or both the first repeat interval and the third repeat interval based on the modification instruction.

33. The method of claim 30, wherein:

the first beacon signal and the second beacon signal are each a first signal using an open protocol for alerting the location logging application to a presence of the locator beacon; and

the first advertising signal is a second signal using a proprietary protocol for providing the identification of the locator beacon.

34. The method of claim 30, wherein the location of the locator beacon is determined based on a location of the wireless device.

35. The method of claim 30, wherein a second location of the locator beacon or another locator beacon is logged by the location logging application in response to the wireless device receiving another first advertising beacon signal from the locator beacon or a second advertising signal from the another locator beacon, and

wherein the second location of the locator beacon or the another locator beacon is not logged until (i) the location logging application returns to a dormant state after being active, (ii) the wireless device receives, when the location logging application is in the dormant state, another of either one of the first beacon signal and the second beacon signal from the locator beacon or another beacon signal from the another locator beacon, and (iii) the wireless device re-activates the location

logging application in response to receiving the another of either one of the first beacon signal and the second beacon signal or the another beacon signal.

36. A locator beacon comprising:

a first wireless transmitter; and

one or more first electronic processors, the one or more first electronic processors configured to:

transmit repeatedly, via the first wireless transmitter, a first beacon signal through a first number of transmission repetitions spaced at a first repeat interval,

transmit repeatedly, via the first wireless transmitter, a first advertising signal through a second number of transmission repetitions at a second repeat interval, wherein the first advertising signal includes an identifier that provides identification of the locator beacon, and

transmit repeatedly, via the first wireless transmitter, a second beacon signal through a third number of transmission repetitions spaced at a third repeat interval, wherein the third repeat interval is different than the first repeat interval, and wherein at least one of the first number of transmission repetitions occurs between at least two of the third number of transmission repetitions;

wherein the first beacon signal and the second beacon signal are each a first signal using an open protocol for alerting a location logging application of a wireless device to a presence of the locator beacon; and

wherein the first advertising signal is a second signal using a proprietary protocol for providing the identifi-

cation of the locator beacon to the wireless device for logging of a location of the locator beacon while the location logging application is active.

37. The locator beacon of claim **36**, wherein the one or more first electronic processors is configured to:

transmit repeatedly, via the first wireless transmitter, a third beacon signal through a fifth number of transmission repetitions spaced at a fifth repeat interval, wherein the fifth repeat interval is different than the first repeat interval and the third repeat interval, and wherein at least one of the third number of transmission repetitions occurs between at least two of the fifth number of transmission repetitions;

wherein the third beacon signal uses the open protocol for alerting the location logging application of the wireless device to the presence of the locator beacon.

38. The locator beacon of claim **36**, wherein the one or more first electronic processors is configured to:

receive, via the first wireless transmitter, a modification instruction from the wireless device or another wireless device; and

modify the first repeat interval, the third repeat interval, or both the first repeat interval and the third repeat interval based on the modification instruction.

39. The locator beacon of claim **36**, wherein the location of the locator beacon is determined based on a location of the wireless device.

40. The locator beacon of claim **36**, wherein the locator beacon is integrated within a power tool.

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